surface	R	G	B	clear	R/W	G/W	B/W
white	6640	5100	2815	15100	1.00	1.00	1.00
red	3960	1420	870	6150	**0.60**	0.28	0.31
green	2340	2630	980	6190	0.35	**0.52**	0.35
blue	760	580	585	1930	0.11	0.11	**0.21**
grey floor	2080	1575	875	4685	0.31	0.31	0.31

Exhibit 10A. What the colour sensor reads on each surface. TCS34725, 100 ms, gain 4, steps 3 and 6, 20 Oct 2026. Raw counts, then each channel divided by its white reading. Raw blue is never the largest, even on the blue stripe, because the LED is weak in blue. Divided by white, each stripe's own channel is the largest, which is how classify() in colour.py names it. The grey floor reads equal ratios.

Fig. 10.1. The steering rule, follow.py steer(), rev f5c0688, following blue at 30 rpm. A cross section of the track as the rover sees it, red on its left. Under each part: left / right wheel rpm. The turn grows with the drift. On blue the inner wheel slows in proportion to green minus red (zero at the centre, read at start). On an edge it runs at half speed or less, on red, green or white it stops.

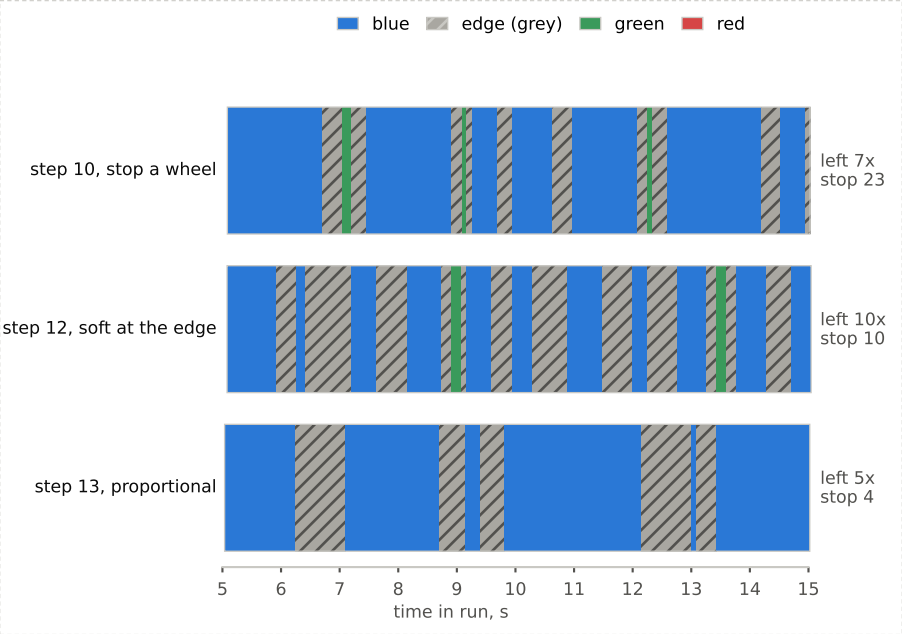

Fig. 10.2. The same straight, 5 to 15 s, under each steering version. One cell per loop, about 85 ms, coloured by what the sensor saw. Steps 10, 12 and 13. Right: times it left blue, and loops with a wheel under 10 rpm. Stopping a wheel at every drift jerks it (23 loops). Soft turns at the edge cut that but it wanders to the edge more often. Proportional steering leaves blue least, never reaches red or green.

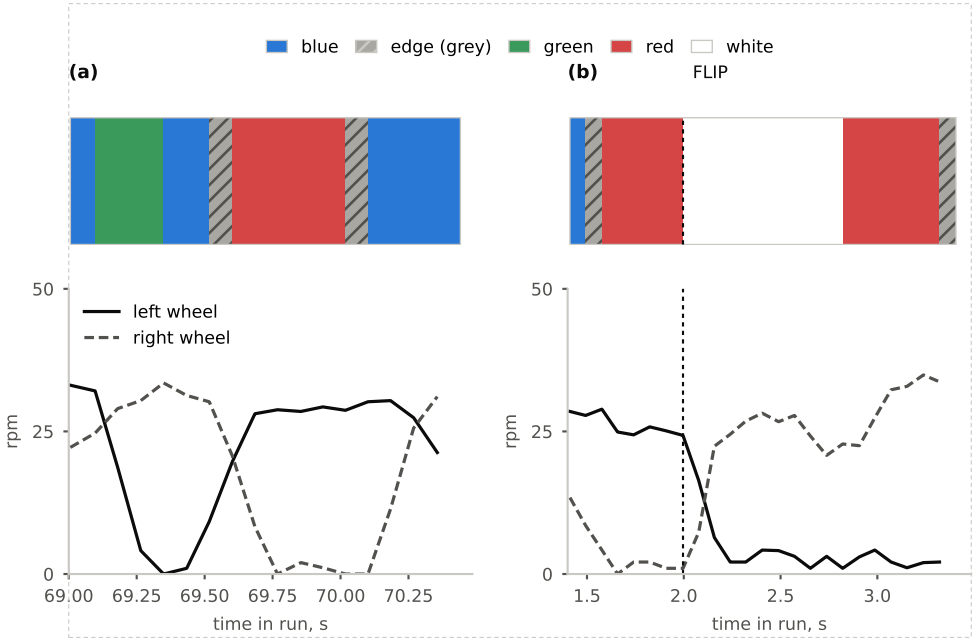

Fig. 10.3. (a) The figure eight crossing, step 14, 69.0 to 70.4 s. The other strand's green, blue and red pass under the sensor in 1 s. Green turns it left, red right, and the two cancel. The other blue lasts 0.2 s, too short to follow. (b) Started the wrong way round, step 15. The turn away from red drives onto white, so the stripe order flips at 2.0 s (dashed line) and it turns back onto blue. Wheel rpm below.